Algebra 1
Unit 8
Lesson 8 Practice

Name Answer Key____________
Date_____________________
Period____________________

The angle, in degrees, at which a batter hits a baseball can affect the distance the ball travels, in feet. The function $B(x) = -0.2x^2 + 14x - 21$ models the distance the baseball travels in ft. depending on the angle, in degrees, at which the ball is hit.

1. What does the variable x represent? Include the units.
   x represents the angles in degrees, at which the ball is hit.

2. What does the variable y represent? Include the units.
   y represents the distance the baseball travels in feet.

3. Interpret the y-intercept. Include units.
   The y-intercept is -21 feet. At an angle of $0°$, the baseball travels -21 feet.

4. Rewrite the function in vertex form. Show your work.
   $-0.2(x^2 - 70x + 1225) - 21 + 245$
   $-0.2(x - 35)^2 + 224$

5. Interpret the vertex in terms of the context. Include units.
   The vertex is $(35,224)$. At an angle of $35°$, the baseball travels 224 feet.

6. Complete the table of values for $M(x)$.

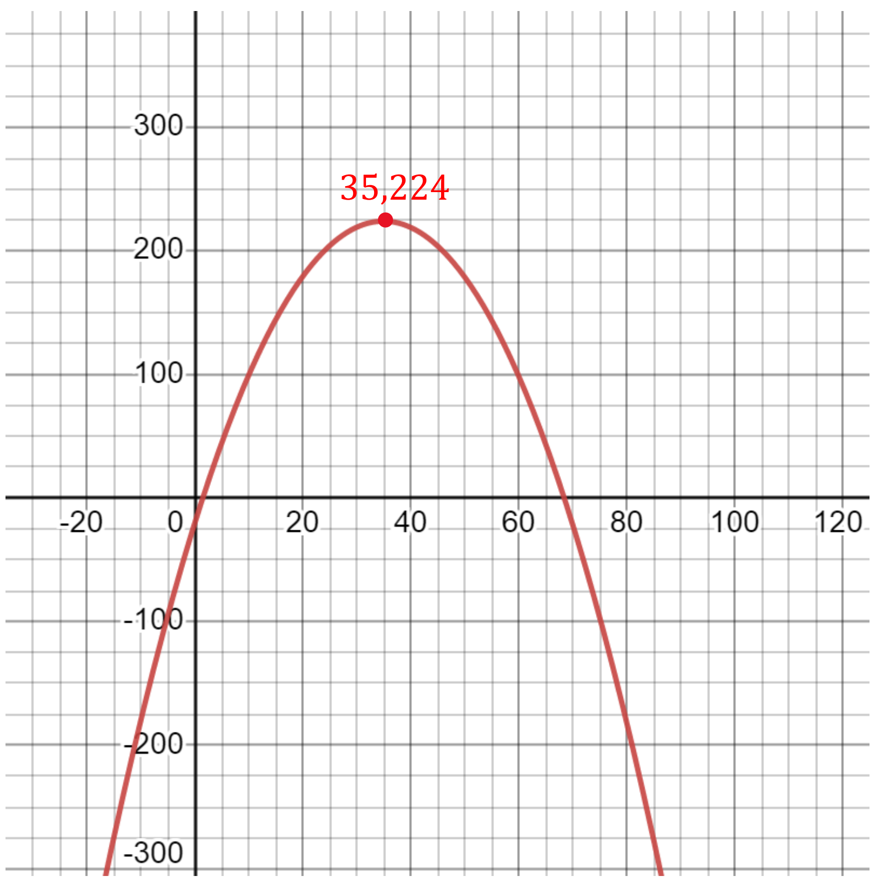
x	-2	0	2	7	17	23	35	68
$M(x)$	-49.8	-21	6.2	67.2	159.2	195.2	224	6.2

7. Circle the values in the table that do not make sense in terms of the context.

8. Complete as many of the key features as possible using the given function, the function rewritten in vertex form, and the table of values. Do not use approximate or rounded values.

Key Features					
x-intercept		vertex	$(35,224)$	domain	$\mathbb{R}$
x-intercept		increasing	$x < 35$	range	$y \leq 224$
y-intercept	$(0,-21)$	decreasing	$x > 35$	opens	Down
left end behavior		$x \to -\infty$, $y \to$ $\underline{-\infty}$			
right end behavior		$x \to \infty$, $y \to$ $\underline{-\infty}$			

9. Graph the function.



10. Use the graph, table of values, or function to determine the reasonable constraints for the context. Write the constraints in set builder notation.
$\{x | 1 \leq x \leq 69\}$ using estimated values from table.